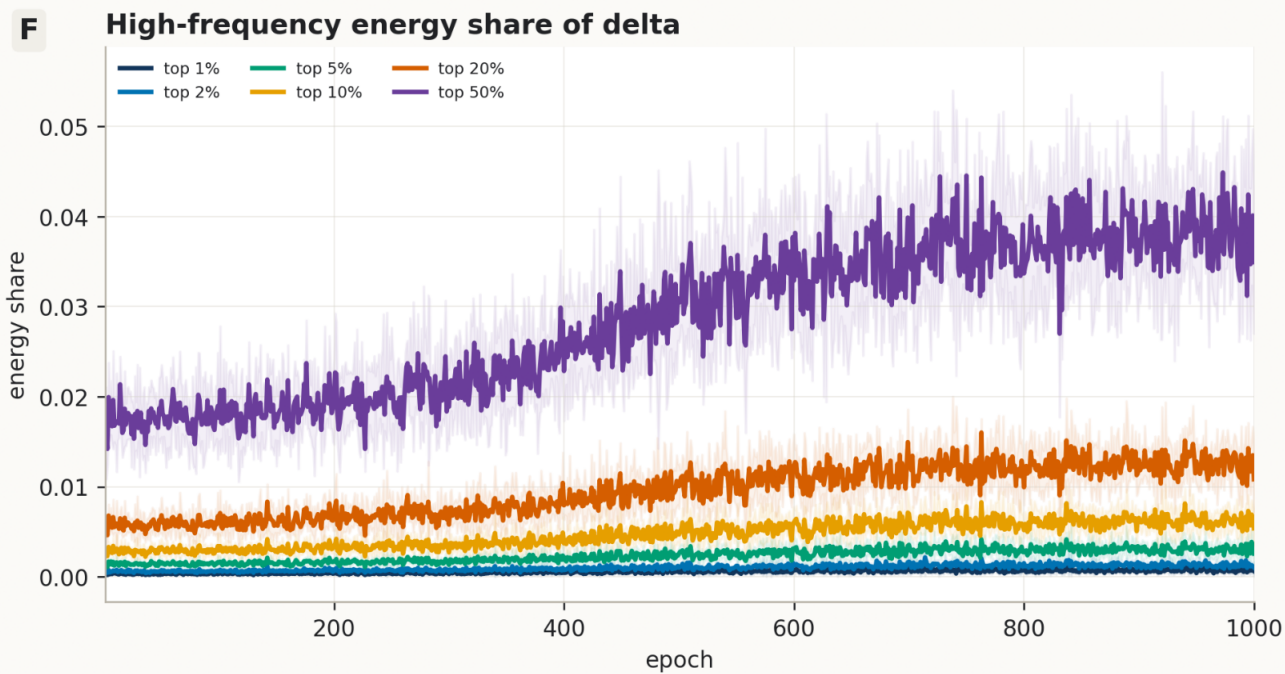
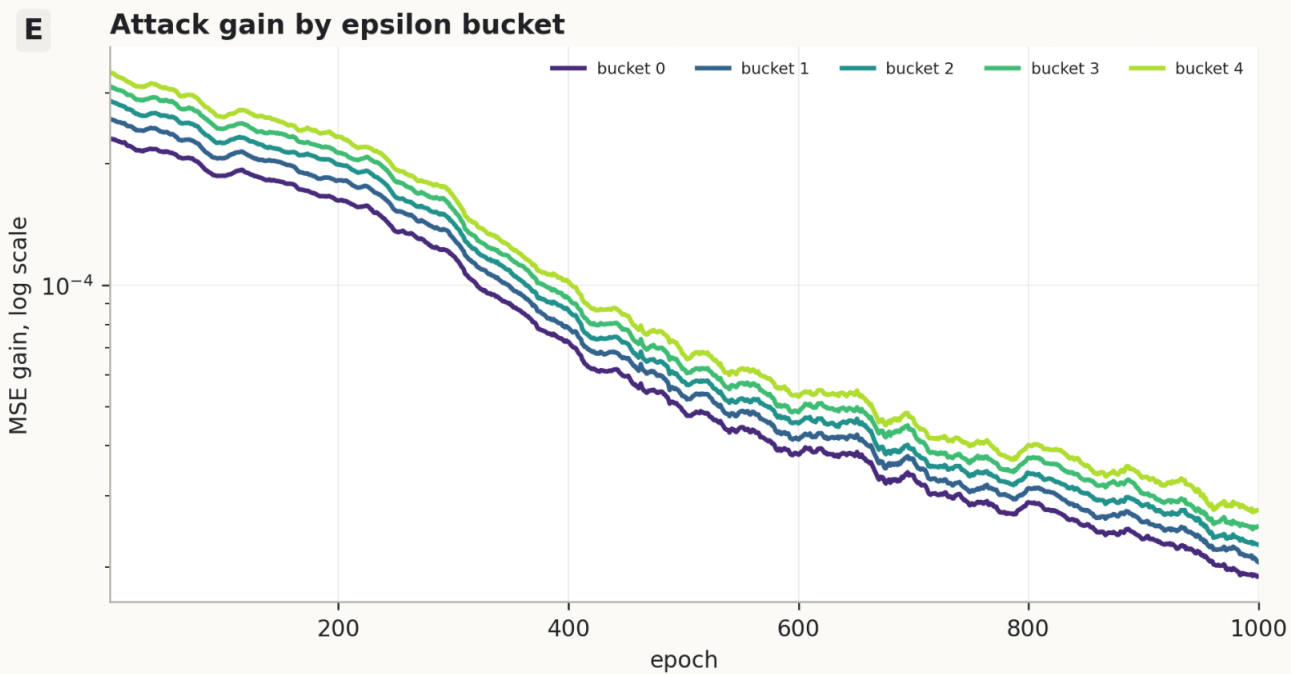
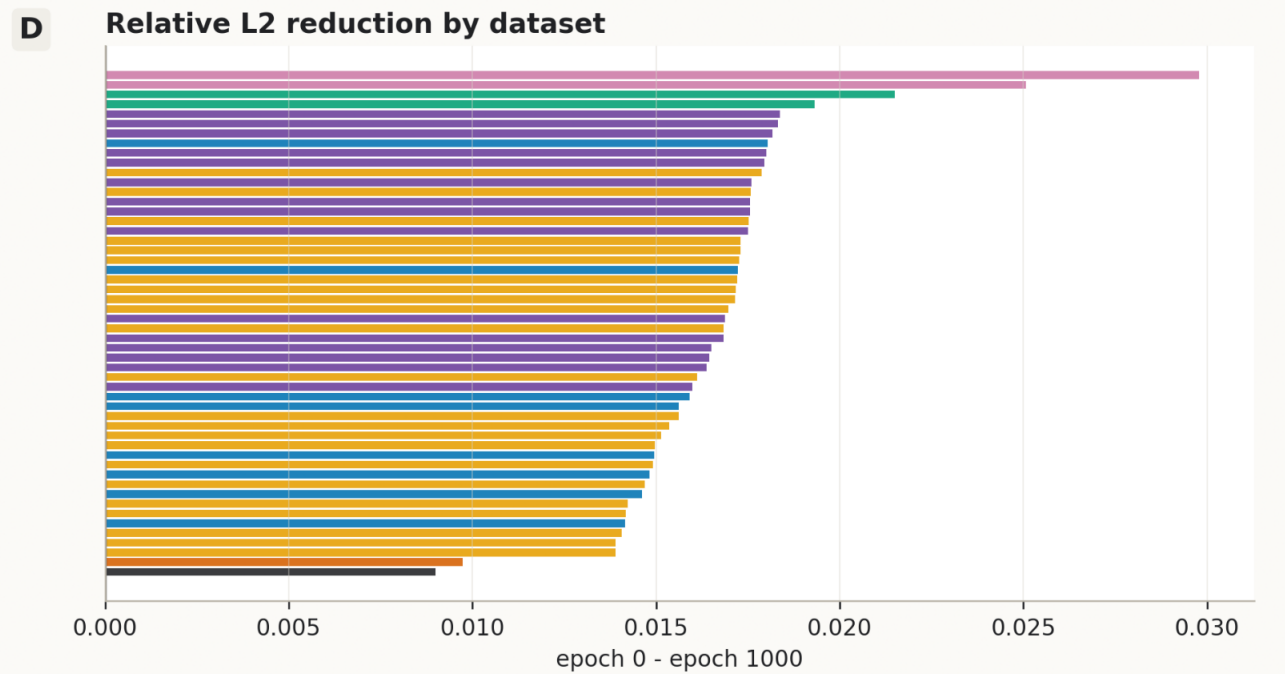
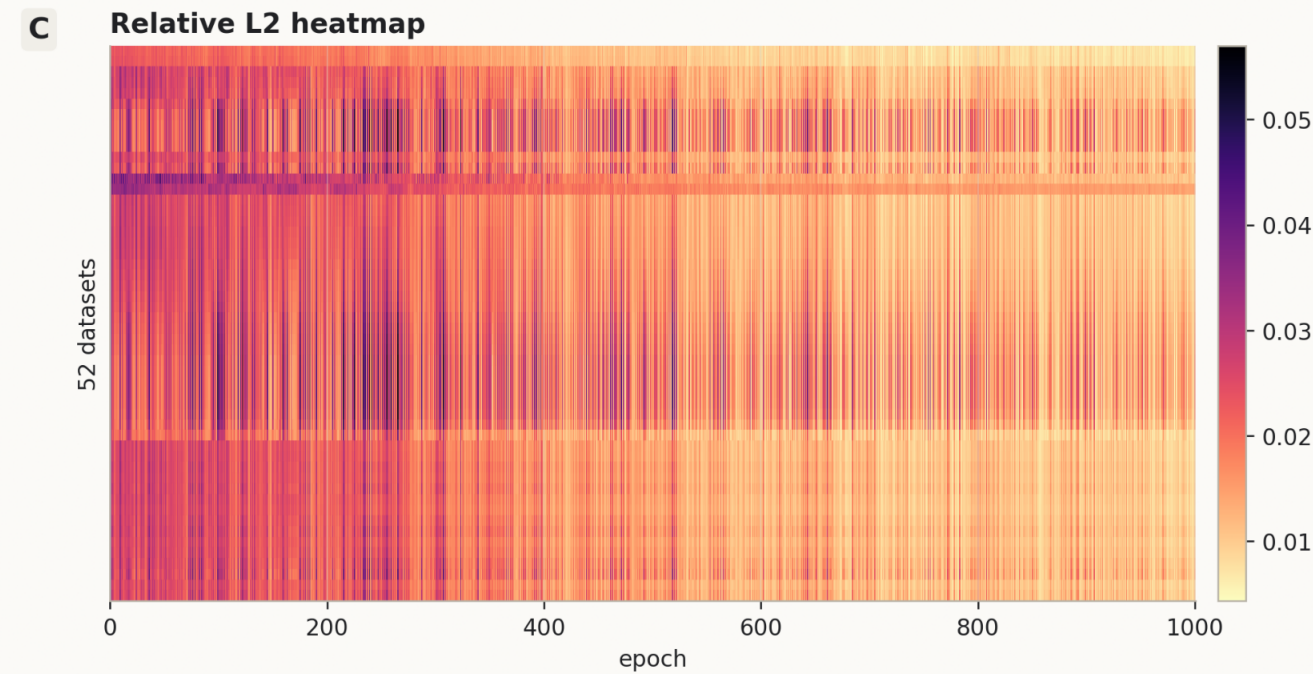
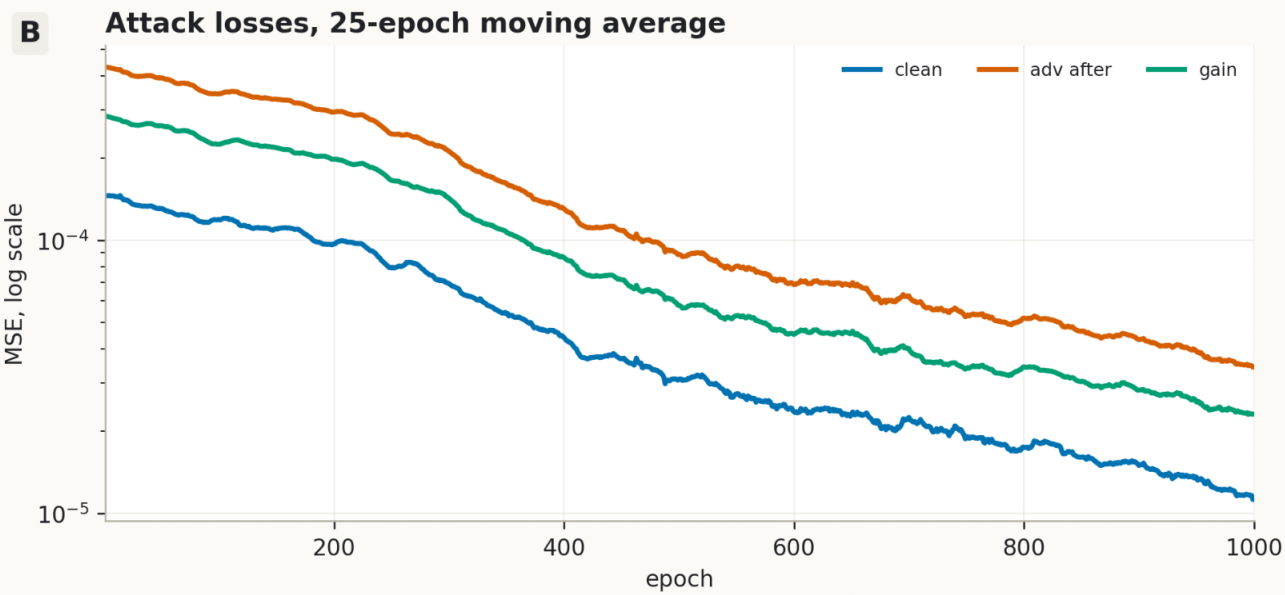
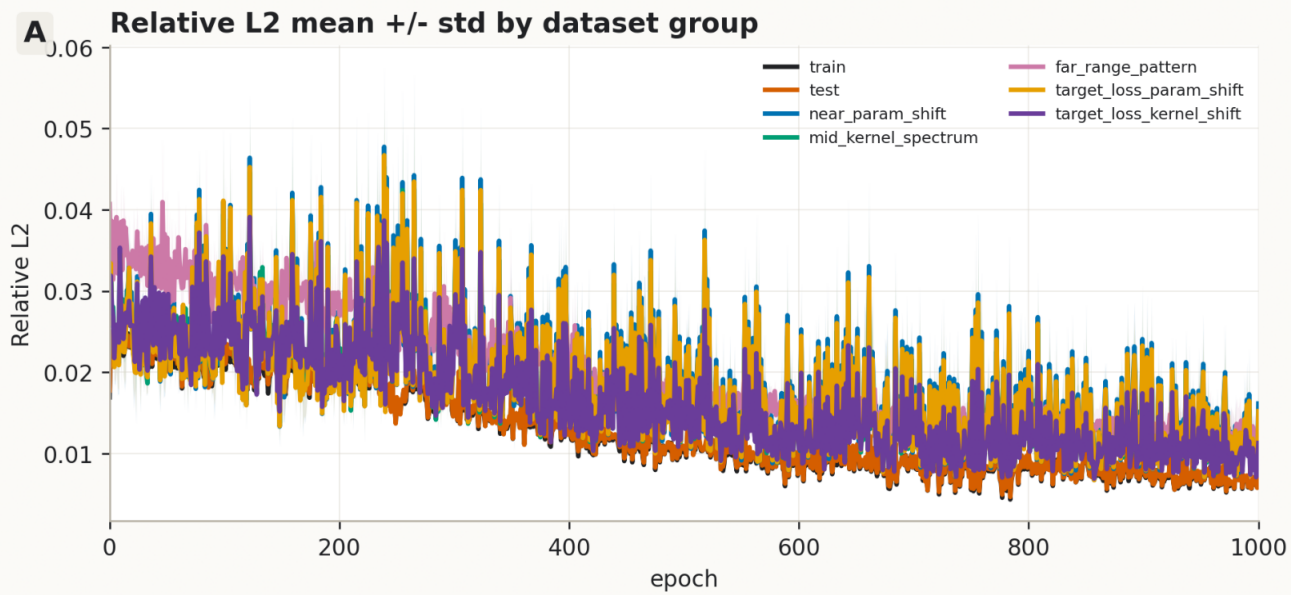
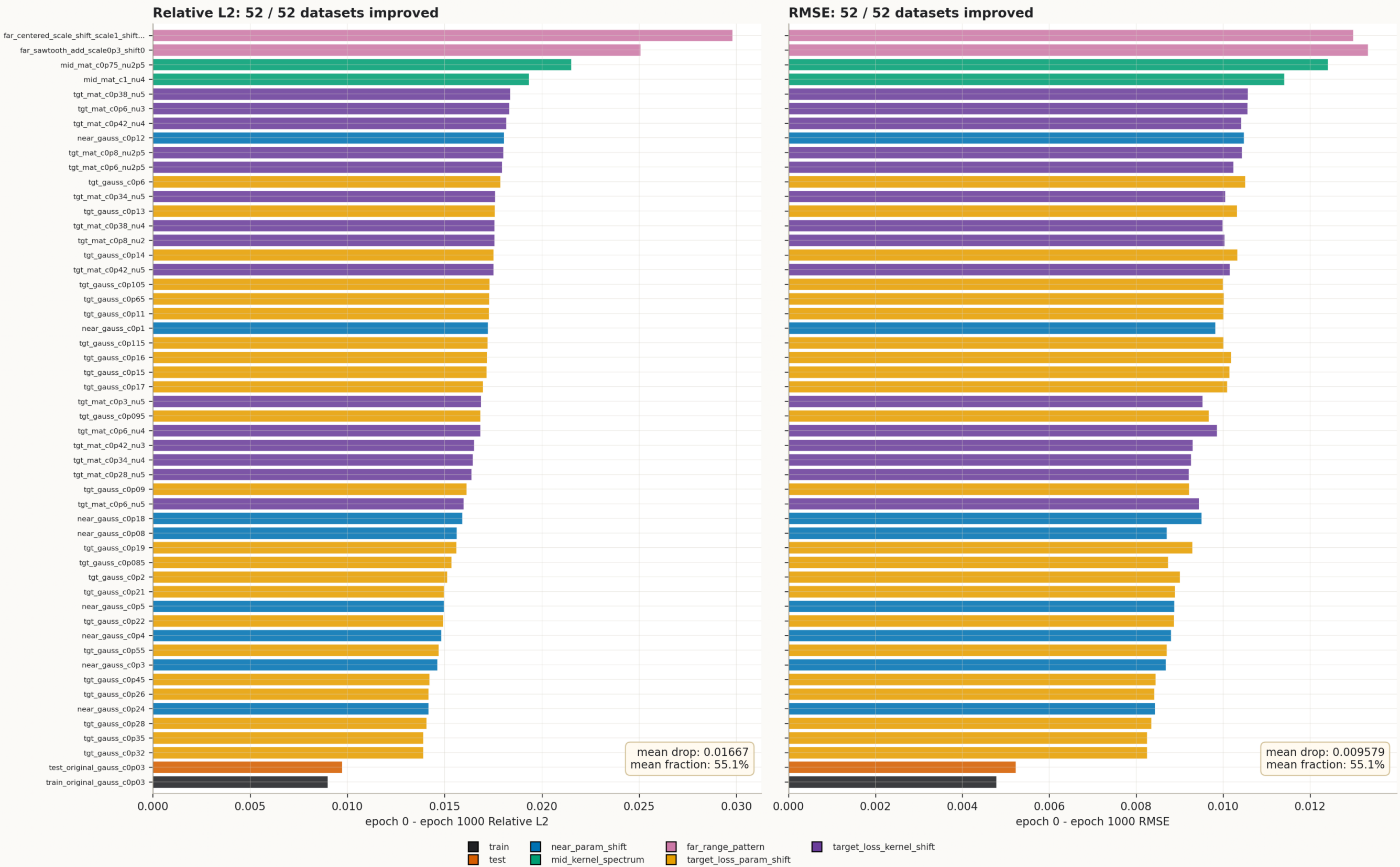


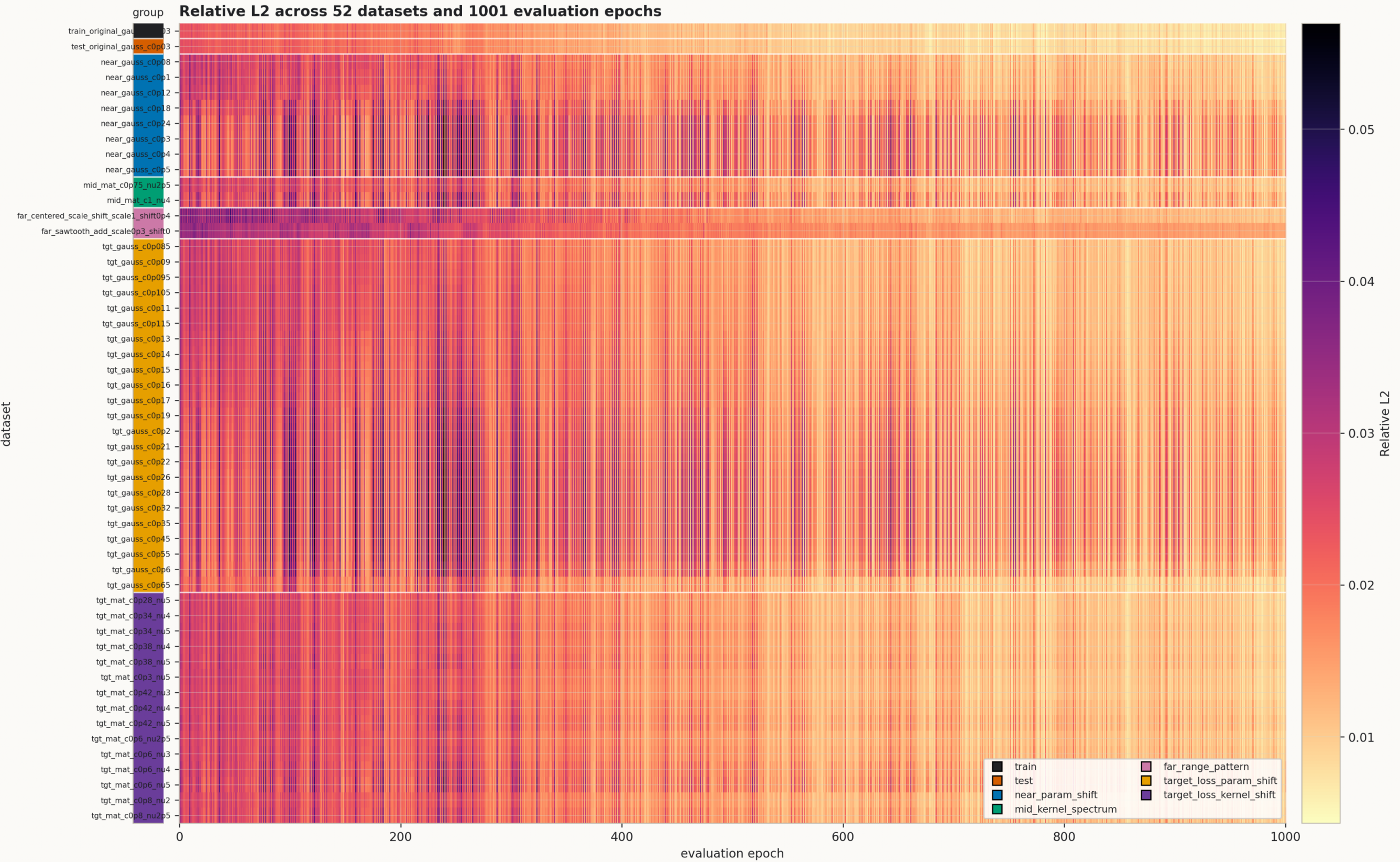
Burgers zero adversarial training: polished diagnostic dashboard

52/52 datasets improved | mean Relative L2 0.0305 -> 0.0138 | mean fractional drop 55.1%



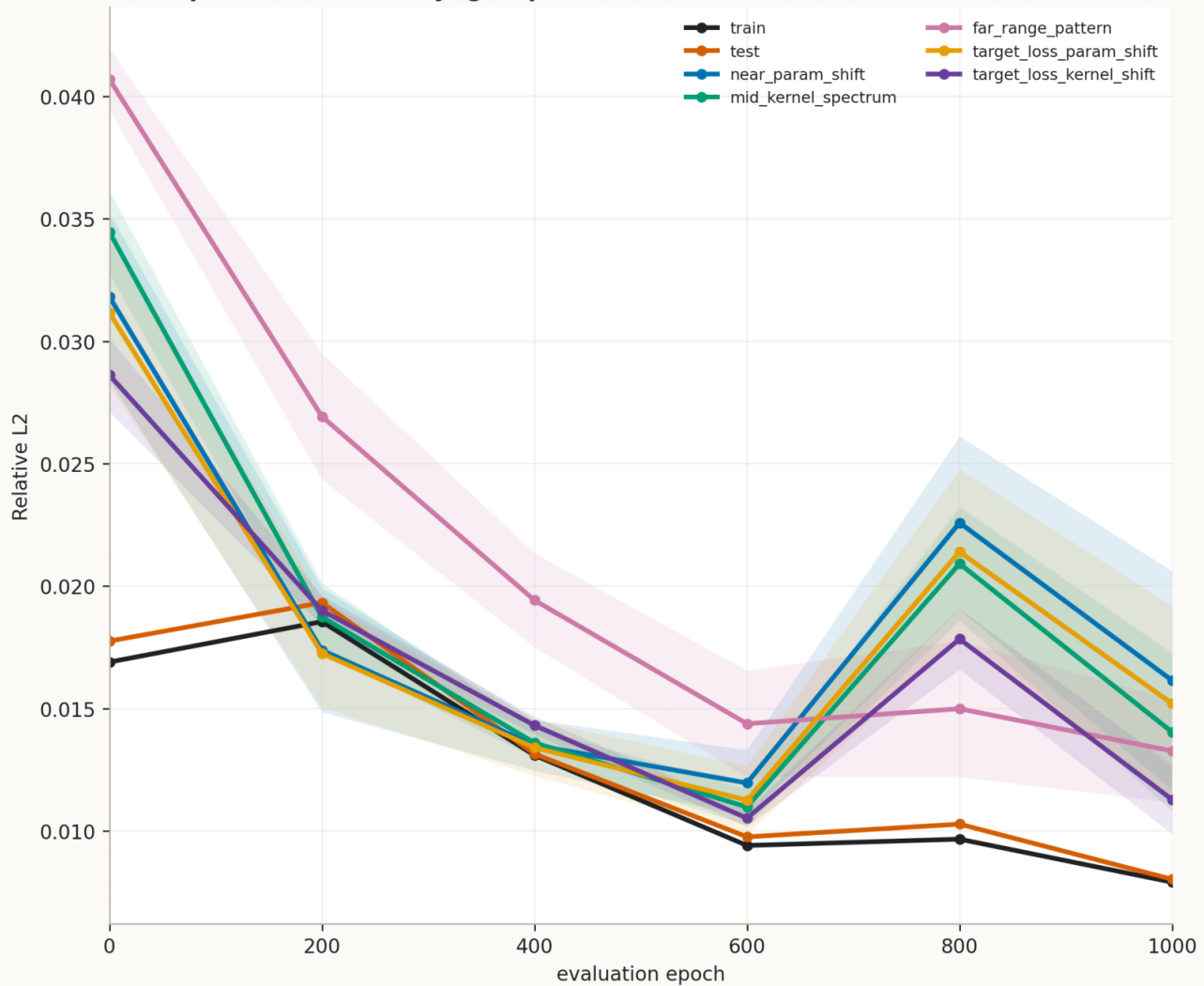
Burgers ADV-only training: loss reduction by dataset



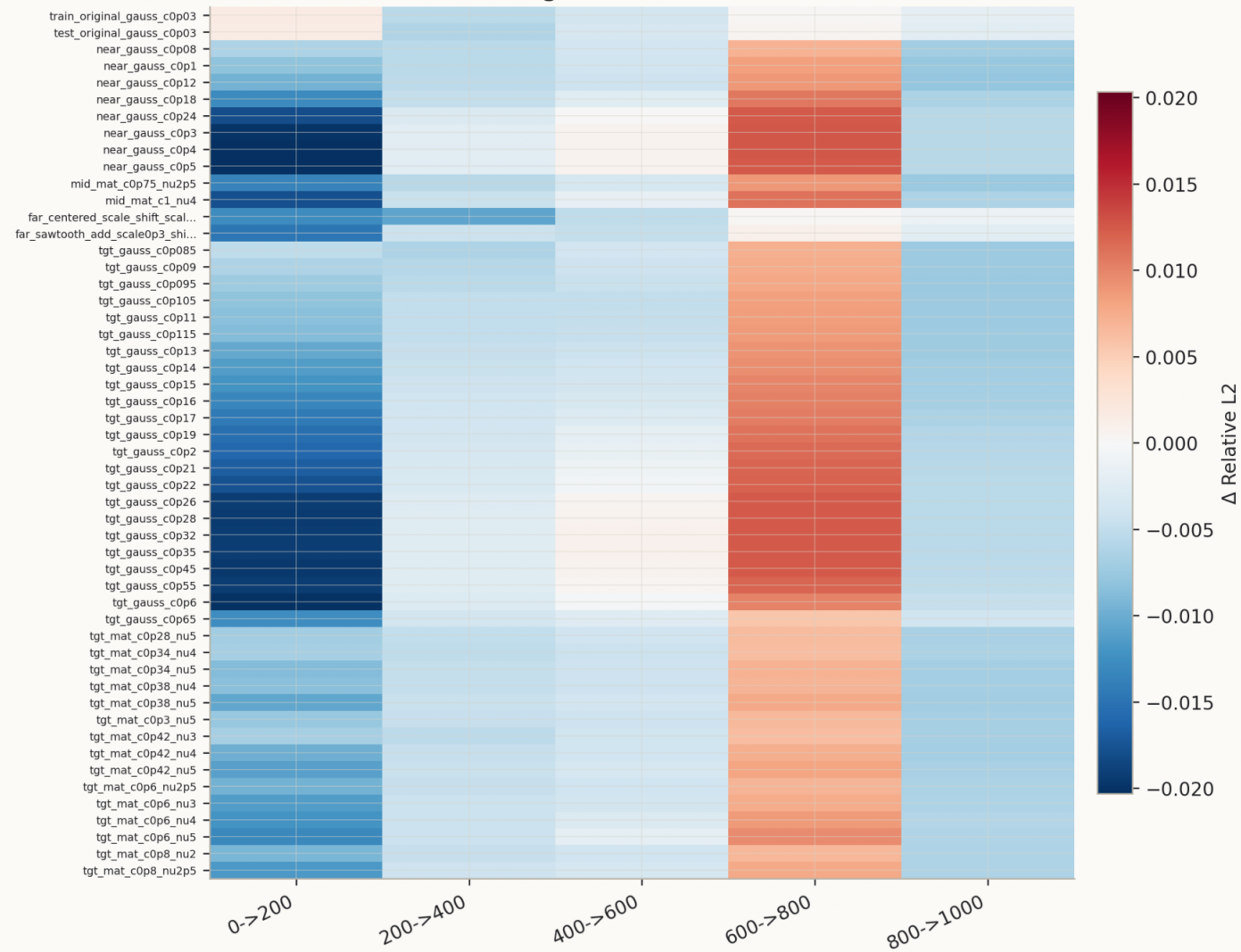


Burgers ADV-only: checkpoint loss behavior

Checkpoint loss summary: group mean $\pm$ std

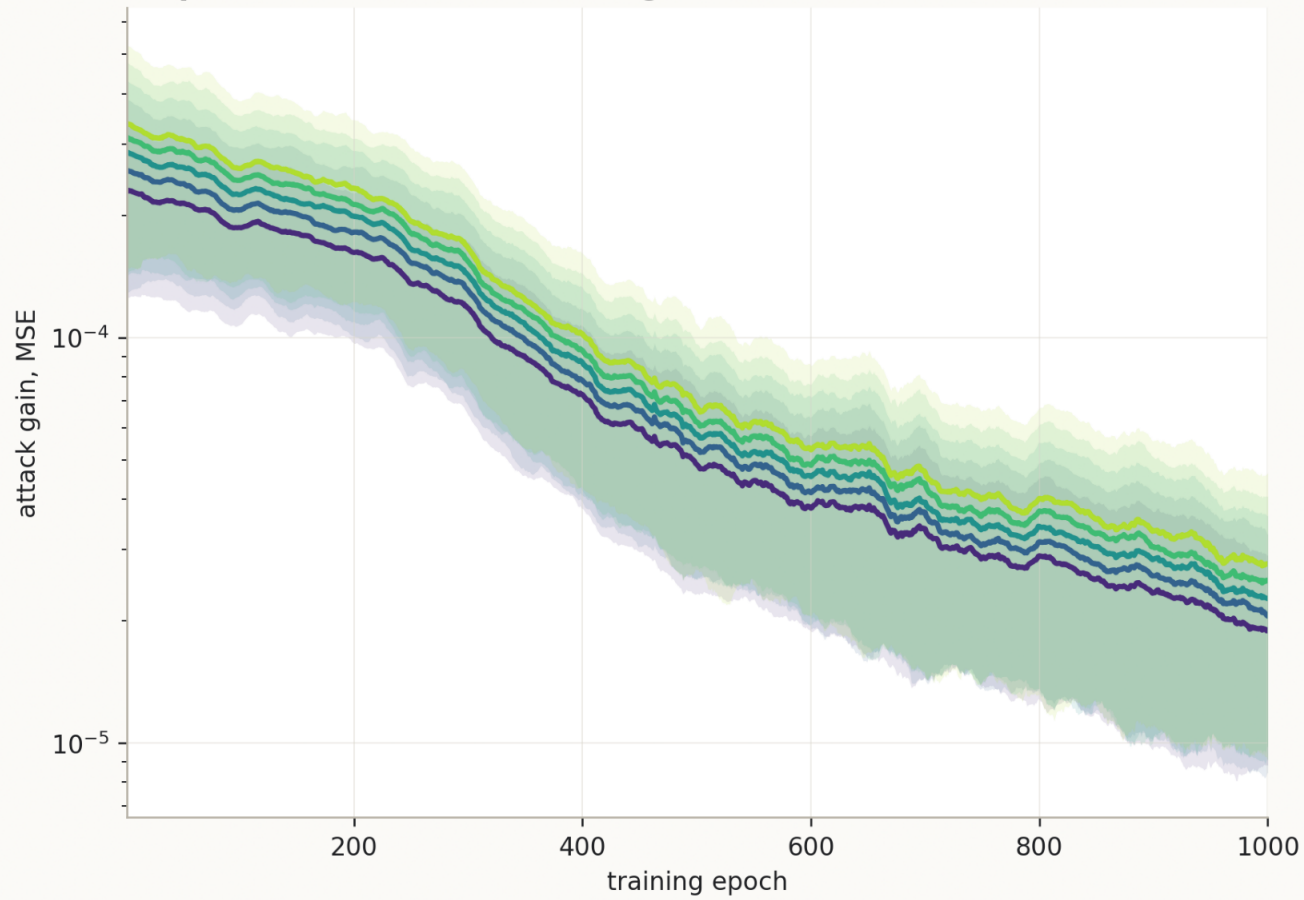


Checkpoint-to-checkpoint change
blue = lower loss, red = higher loss

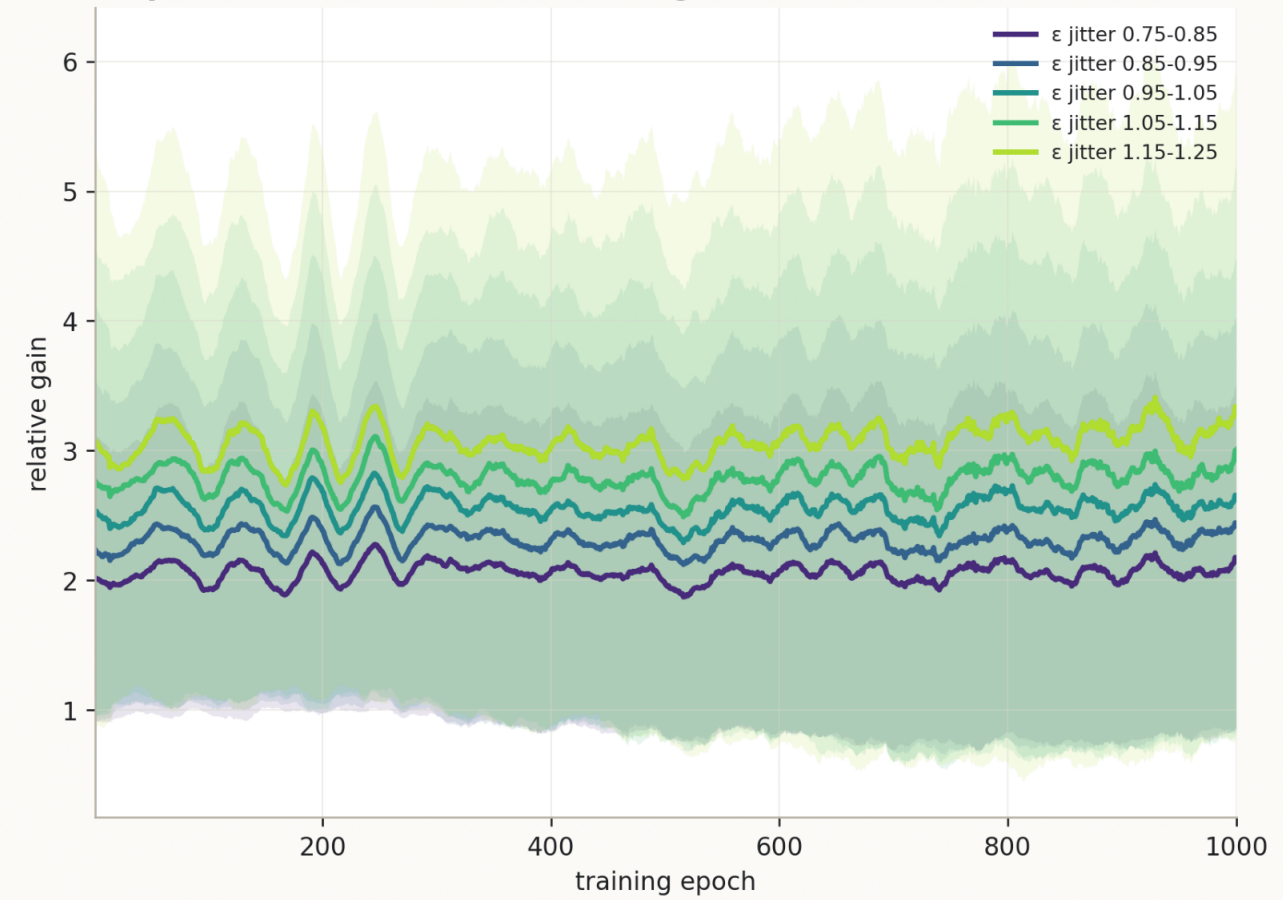


Attack loss bucket analysis

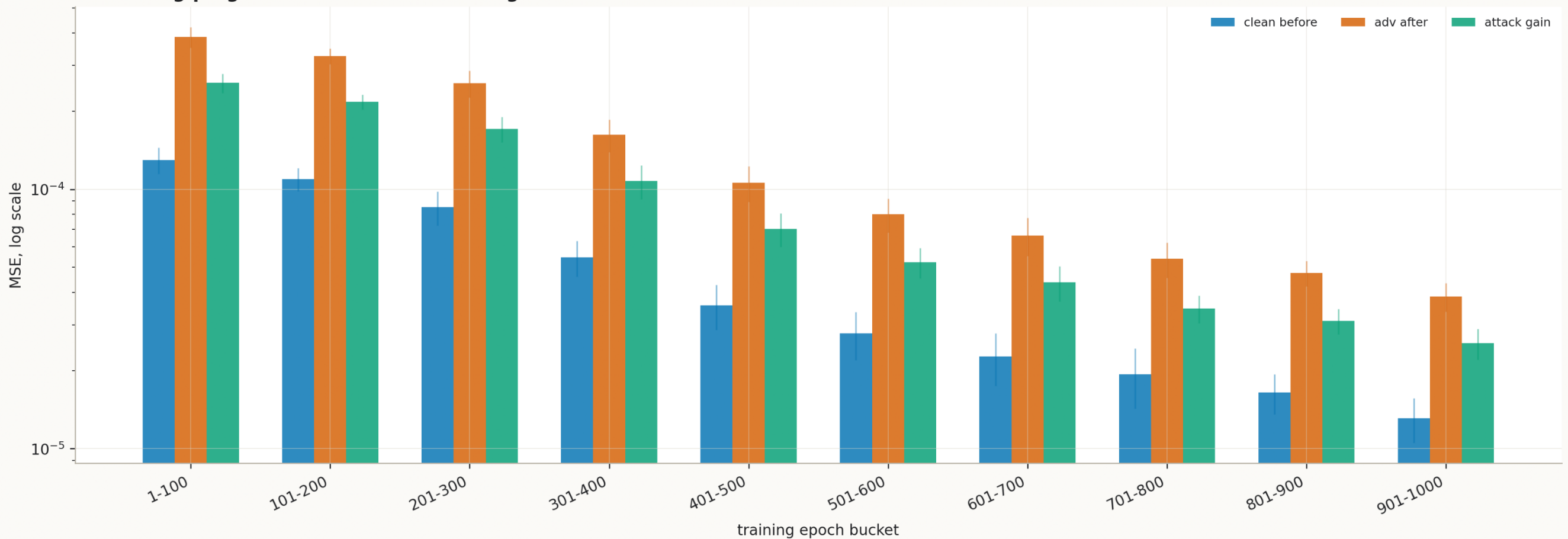
5 epsilon buckets: attack loss gain



5 epsilon buckets: relative attack gain

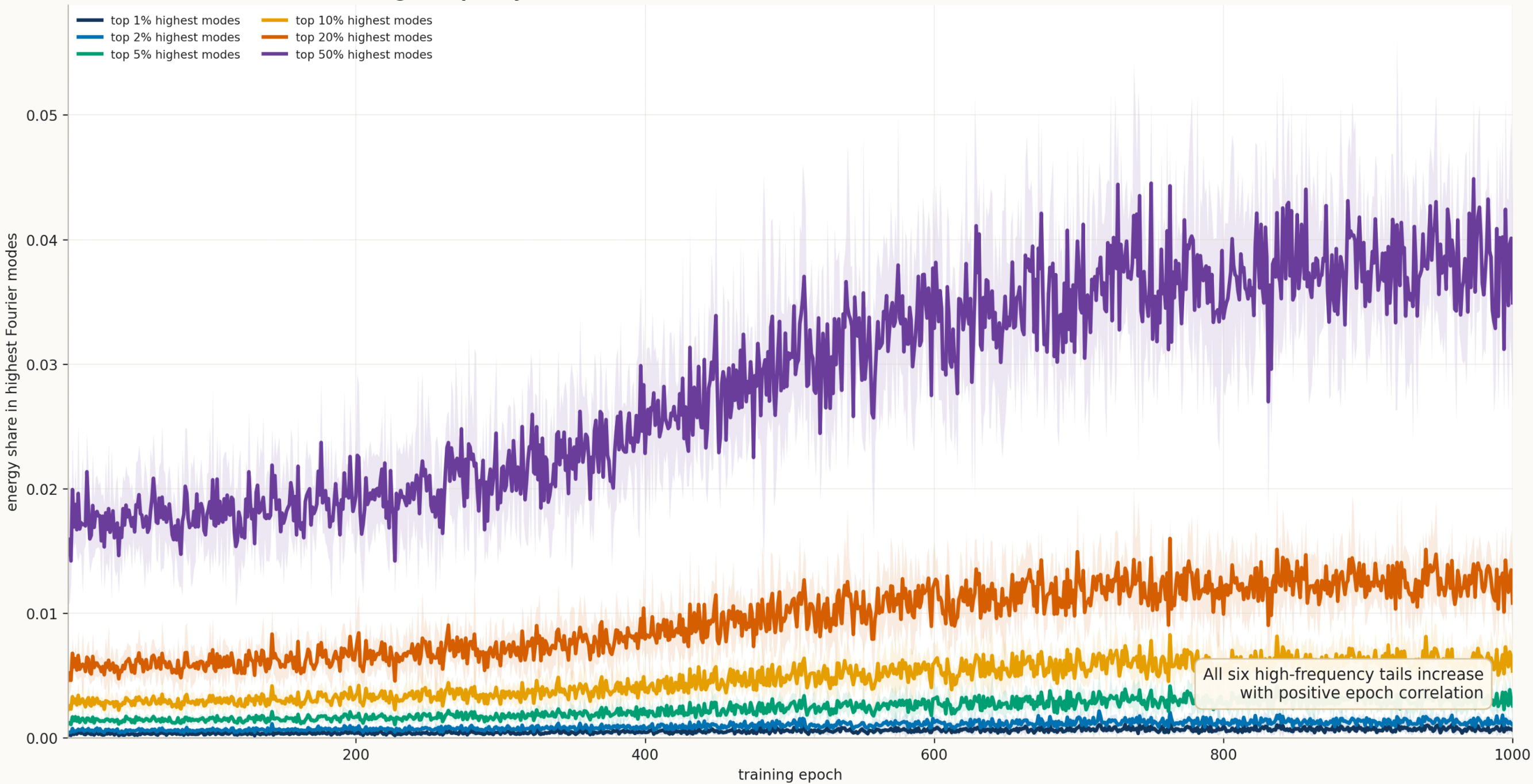


10 training-progress buckets: clean / adv / gain



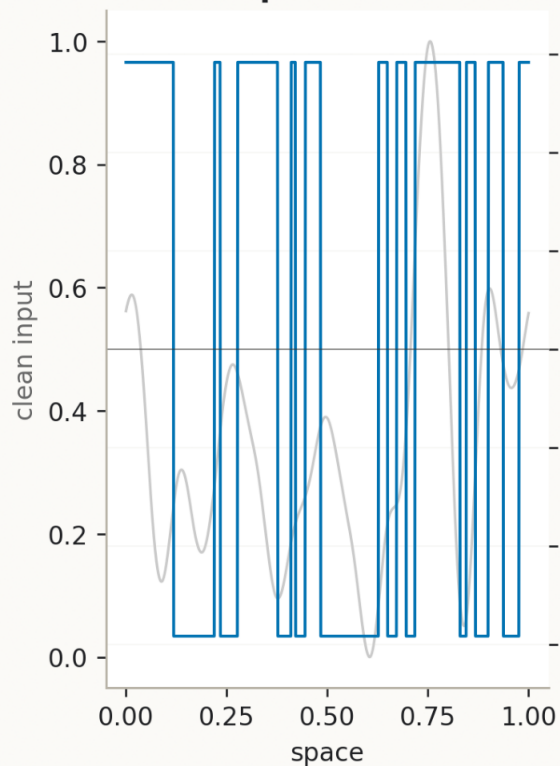
Adversarial delta shifts toward high-frequency Fourier modes

- top 1% highest modes
- top 2% highest modes
- top 5% highest modes
- top 10% highest modes
- top 20% highest modes
- top 50% highest modes

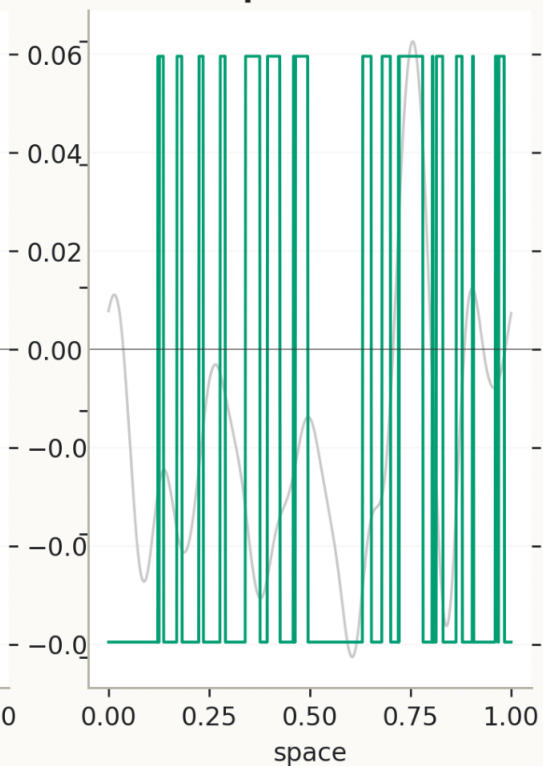


Fixed probe delta checkpoint view: probe 0, source index 0

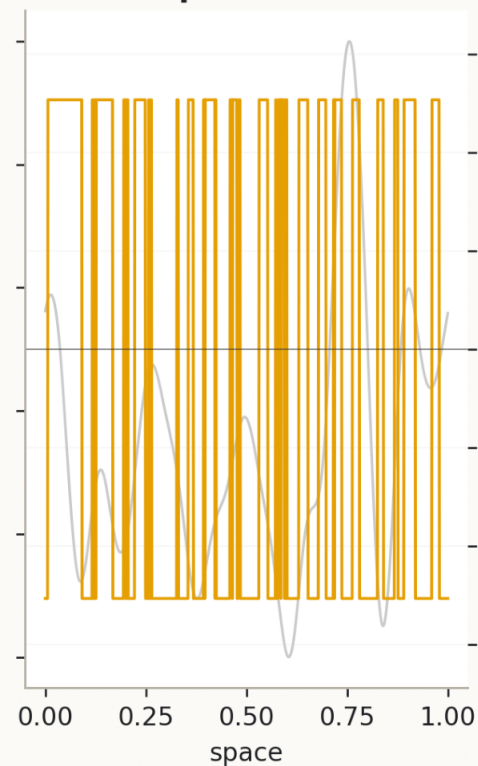
epoch 200



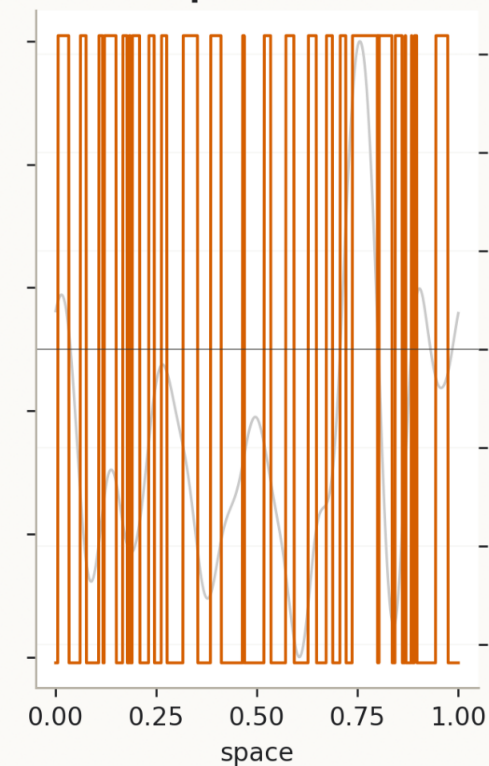
epoch 400



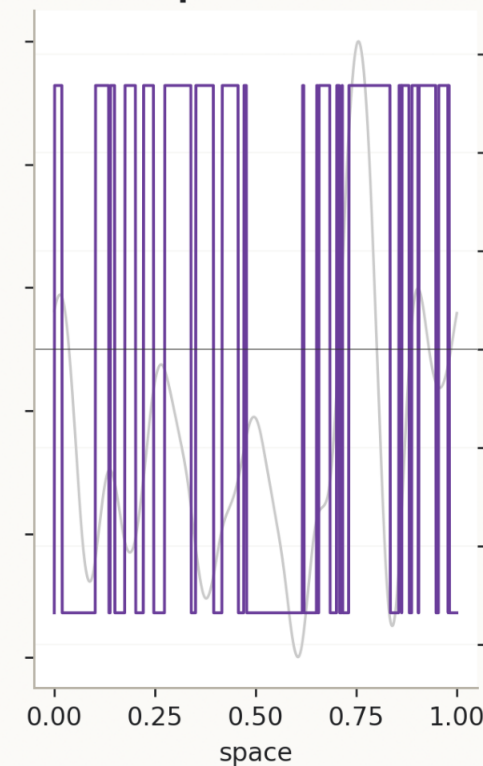
epoch 600



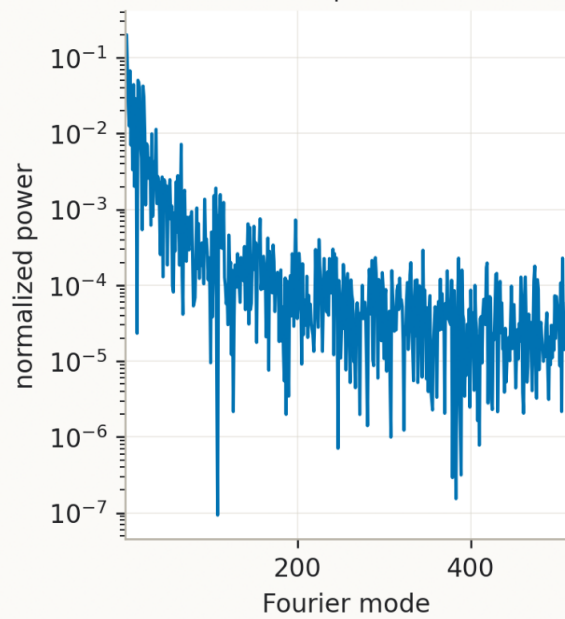
epoch 800



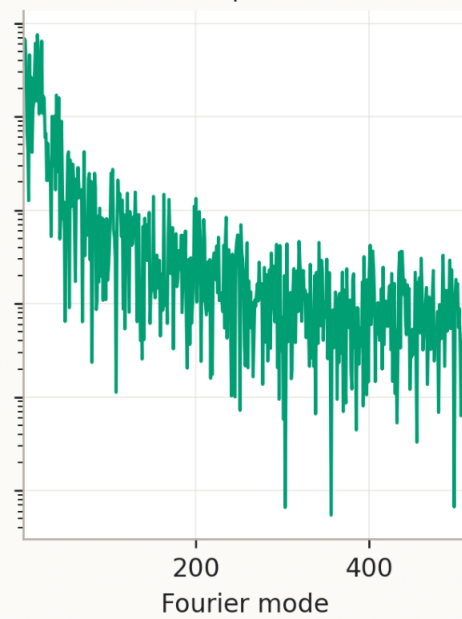
epoch 1000



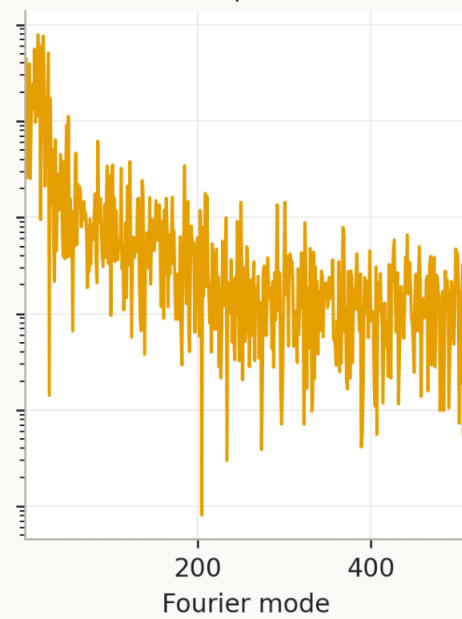
FFT power



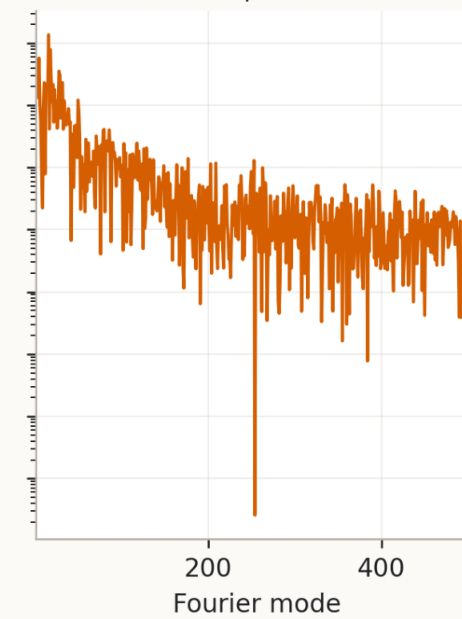
FFT power



FFT power



FFT power



FFT power

